

DHANA KARTHIKEYA VENTRAPRAGADA

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EDUCATION

University of California, Berkeley

Expected Dec 2027

B.A. in Data Science and Applied Mathematics — GPA: 3.85/4.00

Berkeley, CA

- **Relevant Coursework:** Principles & Techniques of Data Science · Probability for Data Science · Data Structures · *In progress:* Introduction to Machine Learning (neural networks, optimization, clustering) · Efficient Algorithms & Intractable Problems · Data Engineering

TECHNICAL SKILLS

Languages: Python, SQL, Java

ML & Statistics: PyTorch, TensorFlow, scikit-learn, XGBoost, LightGBM, SentenceTransformers, embeddings, regression, classification, cross-validation

Data: pandas, NumPy, GeoPandas, Matplotlib, seaborn, PostgreSQL, Jupyter

Engineering: FastAPI, REST APIs, Gemini API, Pydantic, AWS S3, ffmpeg, pytest, Git, GitHub

EXPERIENCE

IDX Exchange | Data Science Intern | Remote

Jun 2026 – Aug 2026

- Reached R^2 0.88 and 8.64% median absolute percentage error (MdAPE) on a held-out month for a California home-price model by benchmarking Linear Regression, Decision Tree, Random Forest, and XGBoost on 71K single-family sales; raising XGBoost depth from 6 to 8 cut MdAPE from 8.89% to 8.64%, a larger gain than lowering the learning rate and doubling the trees (9.01% to 8.89%).
- Recovered a Linear Regression baseline from a dollar-scale R^2 of $-4,647$ to -0.38 by tracing the failure to train/test leakage through worst-prediction error analysis, then recomputing every outlier threshold on training data only.
- Halved Linear Regression MdAPE (29.3% to 14.6%) with a 937-district school feature (spatial join, 5-fold out-of-fold target encoding); retrained XGBoost from scratch and matched R^2 exactly, confirming reproducibility.
- Showed where predictions can be trusted by segmenting error by price quintile (6.5% mid-market, 13% luxury) and 161 school districts, finding district error tracks low sales volume rather than price.

Guang Labs | Engineering Intern | Remote

May 2026 – Aug 2026

- Built a Gemini-based multimodal classification API (FastAPI, PostgreSQL, Auth0) that sorts images and videos into six content domains with confidence and evidence, verified by 86 tests, by running inference as background jobs over four media sources.
- Designed a confidence-gated trust layer that rejects low-confidence, evidence-free, or malformed model outputs and resamples up to 10 fresh clips per long video, with versioned prompts and Pydantic-validated JSON output.

PROJECTS

Movie Genre Classification (Multi-Label NLP) | Python, scikit-learn, SentenceTransformers

- Lifted Macro F1 from 0.48 to 0.66 on 42K+ films across 20+ genres via a rare-genre cutoff and per-label threshold tuning, after benchmarking three embedding models with Logistic Regression, XGBoost, and LightGBM.
- Cut training time $\sim 12\times$ by choosing Logistic Regression on e5 embeddings, which beat both boosted-tree models.

SightLine (Open Projects, ML team) | Multimodal ML, in progress (Fall 2026)

- Contributing to the ML pipeline behind a mobile app that turns a day of Meta Ray-Ban glasses footage into a 60-second recap by splitting footage into clips and ranking them with multimodal audio and visual models.

Housing Price & Assessment Bias | Python, pandas, scikit-learn

- Reached test RMSE 0.562 (4-fold CV) on 200K+ Cook County records with 15+ engineered features, then quantified over- and under-estimation across price bands to show inequities in property-tax assessment.
- Excluded non-arm's-length \$1 sales so the training data mirrored real assessment practice.

LEADERSHIP & INVOLVEMENT

UC Berkeley CDSS | Reader/Tutor, CDSS Seminars | Berkeley, CA

Aug 2026 – Present

- Give resume feedback in Careers in Data Science & Computing; in the Transfer Seminar, run exam reviews for Structure & Interpretation of Computer Programs and Foundations of Data Science and write study-session worksheets.

UC Berkeley EECS | Course Staff (UCS1) | Berkeley, CA

Jun 2026 – Aug 2026

- Taught Structure & Interpretation of Computer Programs: ran 4 weekly exam-prep sections and office hours in Python, SQL, and Scheme, wrote and reviewed exam questions, and built Gradescope autograders.